

AgriDrone AI : AI-Based Multi-Drone System for Smart Agricultural Survey and Crop Disease Detection

**Project report in partial fulfillment of the requirement for the award of the degree of
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CERTIFICATE

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ABSTRACT

The AgriDrone AI system improves the efficiency of farming by using drones to monitor and assess land conditions and crop health. This technology addresses major agricultural problems such as delayed identification of crop diseases, inadequate monitoring of fields, and reliance on manual inspection. AgriDrone AI uses aerial imagery combined with machine learning to provide quicker, more precise, and data-driven evaluations of crop health and land condition, resulting in increased production and decreased waste of resources.

The AgriDrone AI system consists of a manually operated quadcopter drone with a camera used to photograph agricultural fields. These pictures serve as the primary source of information when analyzing and evaluating agricultural fields. This approach differs from a fully automated system in that it offers reliability and cost savings due to the use of human-operated flight combined with automated data processing. The photos taken by the drone are sent to an online system for processing using advanced computer vision and deep learning techniques.

The AgriDrone AI system uses two primary artificial intelligence pipelines. The first one is built specifically for detecting plant diseases. The AI uses a YOLO object detection model to isolate and identify the segmented leaves of plants before performing HSV-filtering on these results to eliminate any background noise found in the image prior to applying CLAHE (constraining the histogram adaptive equalization) techniques (like CLAHE) to preprocess the images such that the resulting visual representation of the plants will be consistent for every image taken even with different levels of illumination in the environment from where the image was produced; Whereas a CNN (convolutional neural network) will then classify the condition of each individual plant as either healthy or diseased and provide the user with a confidence level based on that specific prediction (i.e., the precision of this prediction) before any steps can be recommended for managing that particular plant's health. As a result, this first pipeline allows for timely detection of any issues affecting plant health, enabling managers to quickly address and resolve those issues and reduce the risk of crop loss.

The second pipeline is concerned with detecting field boundaries and providing land segmentation. By using a deep learning edge detection model such as DexiNed, the boundaries between different sections of farmland can be identified using aerial imagery. Once the boundaries have been identified, segmentation techniques will also be applied in order to segment the overall field into smaller plots. After the segmentation has been completed, geometric calculations will be performed to calculate the area of each of the segmented plots from the first step, which can provide valuable insights into land management, crop planning and resource allocation. This will reduce the number of surveys that need to be done manually and improve the accuracy of field analysis.

The overall system is built upon a modern web-based platform that provides the user interface for interacting with and visualizing the system. The frontend is built using technologies such as Next.js and React, which provide functionality for uploading images, performing live detection, visualizing boundaries, and providing analytical dashboards. The backend is implemented with FastAPI, which allows for communication between the machine learning models and provides for efficient processing of tasks. A PostgreSQL database is used to store the metadata related to fields, detections, and outputs from the system, allowing for a historical record of actions completed as well as insights based on data.

AgriDrone AI has many advantages, such as lessening the amount of time needed for agricultural work, speeding up the process of analyzing large areas of agriculture, and increasing the accuracy of identifying crop disease and the boundaries of fields. That would also provide farmers and other stakeholders with real-time data and analysis in order to make better decisions about their crops and other agricultural products, which is part of precision agriculture. While the current system uses manual aerial imagery processed through drones, it is designed to be scalable, allowing for future implementation of features like autonomous drone navigation, video analysis in real-time, Internet of Things (IoT)-based environmental monitoring through sensors embedded in various pieces of equipment, and predictive data models.

In conclusion, AgriDrone AI is a smart agricultural approach that includes data acquisition through drones, data analysis conducted through artificial intelligence, and digital/user-friendly interfaces that help improve how agriculture is monitored today and provide the basis for future development of smart agriculture solutions, leading to improved productivity, sustainable production methods, and a more resilient agriculture industry.

INTRODUCTION

The AgriDrone AI is an advanced agriculture monitor that implements drone technology, A.I. and data-informed decision making tools for modernisation of agri-businesses and traditional agri-practices. Agriculture has been and will undoubtedly continue to be one of the world's top industries providing employment and food security for billions of people around the world. To date, agriculture has continued to face major challenges including, but not limited to, delayed diagnoses of crop health issues, ineffective management of farms/plants/crops, unequal access to all resources, and a heavy reliance on manual labour. These limitations have led to reduced crop yields/production, higher operating expenses, and lengthy time frames to respond to issues affecting crop production. The AgriDrone AI programme will address the aforementioned challenges in agriculture by providing an intelligent, solicitous, and technology-oriented alternative to how we currently monitor, manage, and thereby, overcome challenges related to agricultural production.

The system uses a semi-automated method with human control of the drone for aerial photos of agriculture. This provides a simple, low-cost, dependable solution in cases where a fully autonomous drone is not possible due to regulations, technology, or cost restraints. The drone is used as a source of data collection through a camera, taking high-resolution images from different heights and angles of the crops and field in order to perform extensive analyses of crop conditions and land characteristics through image analysis rather than visiting the field.

A powerful artificial intelligence framework is at the core of AgriDrone AI - it processes captured images using two main analysis pipelines: the plant disease detection pipeline detects plant health problems at an early stage with the use of deep learning and computer vision techniques. The plant disease detection pipeline uses an object detection model based on YOLO (You Only Look Once) to locate and separate the leaf area in an image. This step is important since this helps to filter out irrelevant data such as the ground, the sky or surrounding objects, and ensures that only useful data on the plants is included in the analysis. After this has occurred the regions identified are validated with HSV-based filtering to ensure they represent vegetation and further minimise the number of false positives.

The original content has been paraphrased with the intent of making it consistent with the intent of the original writing.

In order to improve their analysis results, the system will also perform some image preprocessing techniques. The system incorporates image preprocessing techniques like Contrast Limited Adaptive Histogram Equalization (CLAHE) for improving the images' contrast and visibility under different lighting conditions. This is very crucial when working with agricultural and other environments since there may be huge changes in lighting between locations.

After pre-processing the images, the resulting image is passed through a convolutional neural network (CNN) that has been trained on a number of plant disease datasets. Each image is classified by the CNN into either a healthy or a diseased condition, along with a confidence score that indicates how confident the model is about its prediction. This type of automated classifier will allow fast and early detection of diseases, thus enabling growers to take action to prevent the spreading of disease throughout the entire field before they establish themselves in larger numbers.

The second area of focus for this system is the detection of field boundaries and segmentation of the different parts of agricultural fields. There are many times that an agriculture field is split up into various different areas (plots) that all have different uses, but all relate to agriculture, such as rotational crops, irrigation systems, or test plots. It takes a lot of time and is very hard to identify and measure these plots manually. To identify and measure the field boundary, the AgriDrone AI system uses edge detection technologies with advanced models including DexiNed, to detect where the boundaries of differing regions are located in an aerial image. Once the boundary has been detected, the images of the boundaries will be viewed and processed using segmentation algorithms to break up the entire field into individual plots. Using geometric calculations the system will provide an estimation of the area of each plot to allow for accurate measurements of the plots that can be used for planning and monitoring and optimization of the agricultural activity.

In addition to its analysis functions, the AgriDrone AI offers users a complete web-based application that has a centralized user interface. The frontend was built using some of the latest web technologies such as React and Next.js and is mobile-responsive and allows for user interaction. Users can easily upload images, initiate the analysis process, and view their results all from the dashboard in a cohesive manner. The interface will show users key pieces of information such as:

- Total number of plots detected
- Area of individual plot(s)
- Results of disease classification
- Confidence score for disease classification

There are a number of visual representations built into the dashboard which help users to visually interpret complex data. For example, users will receive segmented maps of their agriculture fields with visual highlights indicating where detection occurred.

On the backend, FastAPI is used to facilitate communication between the frontend and the AI models. FastAPI also manages the processing requests for images, executes machine learning inference and then returns the results efficiently. Additionally, a PostgreSQL database is utilized to store system data such as field data, detection results and historical data. This allows users to track their detection activity over time, analyze trends, and provide a structured repository to hold their agricultural data. The modular architecture of the system allows it to expand into new areas, such as real-time monitoring, integrating sensor data from farm machinery and predictive analytics.

While AgriDrone AI enhances efficiency, it also encourages precision agriculture, with decisions made based on reliable timely data instead of assumptions. The system alleviates the need for manual inspection by minimizing time and labor while providing more accurate observations. Through this technology, farmers are able to survey vast tracts of land rapidly; identify areas requiring attention; allocate resources appropriately based on the level of need; identify early symptoms of disease help mitigate crop failure and reduce the use of unnecessary amounts of pesticides, therefore re-enforcing sustainable farming practices.

In addition, the system establishes a strong basis for future smart agricultural advancements. At present, the drones are operated manually, and the architectural design reflects a future transition toward the use of autonomous drones, coordination amongst multiple drones, and analysis of video feeds as they are being transmitted in real-time. Future enhancements of the system will include integrating IoT (Internet of Things) sensors to provide environmental monitoring of soil moisture, temperature, humidity, etc. and thereby provide a greater comprehensive picture of current field conditions. Lastly, the incorporation of advanced analytics along with machine-learning models will enable predicting yields, identifying nutrient deficiencies, and developing optimal farming strategies.

Overall, AgriDrone AI is a holistic, effective way of addressing the issues that exist between traditional farming methods and advanced technology. With drone data collection, the use of artificial intelligence to analyse data, and a user-friendly web interface, AgriDrone AI provides users with the necessary tools to monitor the state of their crops, evaluate their property's layout and manage their land more effectively. AgriDrone AI also provides solutions to many of today's key challenges within the agricultural sector such as late disease detection, misuse of resources, and reliance upon manual inspections by improving productivity and reducing labour time.

Moreover, AgriDrone AI helps advance precision agriculture by providing farmers with the ability to make decisions based on data, so that they may discover problems early enough to correct them. AgriDrone AI's modularity and scalability will allow the system to be enhanced over time with features such as integrated IoT sensors, real-time monitoring, predictive analytics, and the capability to use drones autonomously. In turn, these enhancements will increase accuracy, reduce human dependency, and optimise agricultural methods. In summary, AgriDrone AI improves existing agricultural processes and provides a strong step towards building a technologically advanced, sustainable, intelligent agricultural ecosystem, with the ability to provide enough food for the future population of the world.

LITERATURE SURVEY

1. UAV-Based Crop Monitoring and Agricultural Drones

The application of unmanned aerial vehicles (UAVs) in precision agriculture has grown significantly over the past decade, driven by the need for scalable, high-resolution, and on-demand crop surveillance that satellite remote sensing cannot economically provide. Early UAV-based agricultural studies focused on multispectral imaging to compute vegetation indices such as NDVI (Normalized Difference Vegetation Index), which serves as a proxy for canopy health and photosynthetic activity. These initial deployments demonstrated that drone-derived NDVI maps could identify stressed crop zones at sub-meter resolution, enabling targeted intervention over specific field patches rather than broad-acre chemical treatment.

Sa et al. [1] developed an early hybrid UAV platform combining a Pixhawk-based multirotor with onboard ROS processing and cloud offloading for fruit detection tasks, achieving 91% average precision. Their work established the viability of coupling the Pixhawk ecosystem with cloud inference but highlighted a critical limitation: heavy onboard GPU hardware increased UAV weight beyond the optimal endurance operating point for survey missions. This trade-off between computational capability and flight endurance is a persistent challenge in the literature that the AgriDrone platform resolves by fully offloading all AI inference to cloud infrastructure, keeping the airborne payload limited to the imaging and telemetry subsystems.

Bah et al. [3] employed a fixed-wing UAV with an ArduPlane controller for weed detection using R-CNN, achieving 86% precision on row-crop fields. While the fixed-wing platform provided extended endurance (~45 minutes), it fundamentally lacked the hovering capability necessary for low-altitude, close-range leaf-level imaging. The S500 multirotor frame selected for AgriDrone, equipped with 935KV T-Max brushless motors and 1045 propellers, addresses this limitation by supporting stable hover at 3–10 meters altitude, enabling the camera payload to capture leaf-resolution imagery suitable for CNN-based disease classification. The 35A Syclone ESC provides precise motor speed control essential for stable hover in light wind conditions.

Deng et al. [2] deployed a DJI Phantom 4 for real-time pest detection using YOLOv3, achieving 88% mAP at near-real-time inference speeds. Their study demonstrated the practical utility of YOLO-family detectors in aerial agricultural contexts but was constrained by the closed proprietary hardware of the DJI platform, which limits telemetry access, payload customization, and integration with open-source flight management software. The Pixhawk 2.4.8 flight controller employed in AgriDrone AI AI AI AI exposes full MAVLink telemetry over both serial and UDP, enabling the cloud backend to receive real-time position, altitude, battery, and attitude data, which is persisted in the Neon PostgreSQL sensor_data table alongside AI detection records.

Zhang et al. [5] combined a multirotor UAV platform with distributed IoT soil sensors to achieve comprehensive crop stress monitoring, demonstrating that geospatially registered multi-modal data significantly improves the diagnostic precision of disease localization. Their system architecture, however, lacked a real-time cloud dashboard, relying instead on post-mission batch processing.

The AgriDrone AI platform extends this multi-modal vision through its sensor_data schema, which records temperature, humidity, soil moisture, NDVI, battery level, and GPS coordinates at each sampling point, all accessible in real-time through the Next.js dashboard via the /api/sensor-data endpoint with SWR-based polling.

Rehman et al. [6] demonstrated thermal imaging-based disease mapping using an expensive octocopter with a DJI A3 flight controller, achieving high spatial accuracy but at a hardware cost exceeding \$15,000. The AgriDrone AI hardware bill of materials — S500 frame, Pixhawk 2.4.8, M8N GPS, 935KV motors, Syclone ESCs, PM02 power module, 4S 5200mAh LiPo, and iMax B6 charger — represents a field-deployable research platform accessible at a fraction of this cost, while maintaining compatibility with the full ArduCopter/PX4 software ecosystem and the MAVLink ground control protocol.

The M8N GPS module paired with the Pixhawk 2.4.8 provides sub-2-meter position accuracy under open-sky conditions, which is sufficient for waypoint-based field survey missions at standard agricultural survey altitudes of 30–80 meters. The T10 transmitter and R10 receiver pair provides 10-channel PWM control for manual override, a regulatory requirement in most jurisdictions for beyond-visual-line-of-sight (BVLOS) exemptions. The PM02 power module provides voltage and current sensing for battery state estimation, feeding battery level data directly into the AgriDrone AI drone entity schema for real-time fleet battery monitoring through the dashboard.

Table 1. Comparative Analysis of UAV Hardware Platforms for Agricultural Monitoring

Reference	Platform Type	Flight Controller	Sensors / Camera	Endurance	Key Contribution & Limitation
AgriDrone AI (Proposed)	S500 Multirotor	Pixhawk 2.4.8	RGB Cam, M8N GPS, PM02	~22 min (4S 5200mAh)	Open-source cloud-first agricultural platform; full AI offload to cloud
Sa et al. [1]	Custom Multirotor	Pixhawk PX4	Multispectral + RGB	~18 min	Fruit detection via CNN; heavy onboard compute required
Deng et al. [2]	DJI Phantom 4	Proprietary	Multispectral	~28 min	YOLOv3 pest detection; closed hardware, no customization
Bah et al. [3]	Fixed-Wing UAV	ArduPlane	RGB Camera	~45 min	R-CNN weed detection; no hovering for close-range imaging
Dyrmann et al. [4]	Ground Robot	N/A (ground)	RGB Camera	Unlimited	Weed classification 86.2%; terrain-limited, no aerial coverage
Zhang et al. [5]	Multirotor + IoT	Pixhawk APM	Multispectral + Soil	~20 min	Crop stress via UAV + IoT; no real-time cloud dashboard
Rehman et al. [6]	Octocopter	DJI A3	Thermal + RGB	~15 min	Disease thermal mapping; expensive closed hardware stack
Huang et al. [7]	Fixed-Wing UAV	Autopilot Custom	Hyperspectral	~60 min	Nutrient deficiency detection; high cost, not field-accessible

2. Deep Learning Models for Plant Disease Detection

The use of deep convolutional neural networks for automated plant disease detection was established at scale by Mohanty et al. [8], whose landmark study applied AlexNet and GoogLeNet to the PlantVillage dataset of 54,306 images spanning 26 disease classes across 14 crop species, achieving 99.35% classification accuracy under controlled single-leaf, white-background conditions. This result set the benchmark for CNN-based plant pathology classification but also exposed its most significant limitation: a domain gap of 30–40% accuracy drop when identical models were applied to in-situ field imagery captured under variable lighting, background clutter, and partial occlusion. This gap between controlled-dataset performance and real-world field performance remains one of the central unsolved challenges in agricultural AI.

Ferentinos et al. [9] extended the CNN plant disease classification paradigm to drone-captured imagery, achieving 96.3% accuracy on a curated aerial leaf dataset. Their work confirmed that CNN architectures can generalize to aerial perspectives but operated in offline batch-processing mode without cloud deployment. The AgriDrone AI platform addresses this deployment gap by pairing YOLO-based leaf detection with a PlantDiseaseCNN classifier, both served as microservices through a Python FastAPI backend. The YOLO detection stage isolates leaf regions-of-interest from the full aerial frame before passing them to the classifier, substantially reducing the false positive rate caused by soil patches, water bodies, and background vegetation that contaminate raw frame classification.

Deng et al. [2] established the effectiveness of YOLO-family single-stage detectors in aerial agricultural contexts, achieving 88% mAP on pest detection with YOLOv3 at near-real-time inference speeds. Single-stage detectors are particularly well-suited to the AgriDrone AI cloud inference pipeline because their unified regression-based detection eliminates the region proposal stage present in two-stage architectures (e.g., Faster R-CNN), reducing per-image latency from 5–10 seconds to 1–3 seconds on cloud GPU instances. This latency target aligns with the AgriDrone AI detection benchmark of 2–5 seconds per image, which is acceptable for mission-time disease flagging workflows.

Tran et al. [11] demonstrated a CNN-SVM hybrid architecture achieving 95.48% accuracy on a rice disease dataset, highlighting that combining deep feature extraction with classical classification heads can yield competitive accuracy on small, domain-specific datasets where pure end-to-end CNNs may overfit. This finding is relevant to the PlantDiseaseCNN component of AgriDrone, which employs transfer learning from an ImageNet-pretrained backbone to compensate for limited labeled agricultural training data in field collection scenarios.

Rangarajan et al. [12] applied AlexNet and VGG16 transfer learning to tomato disease classification, achieving 91.2% accuracy and demonstrating that deeper pre-trained architectures provide diminishing returns beyond VGG-depth for single-crop disease classification tasks. The DexiNed component of the AgriDrone AI pipeline serves a fundamentally different function from disease classification: it performs dense edge detection for field boundary delineation in the aerial imagery, enabling the system to segment treatment zones and calculate affected area in square meters for the `plant_detections` schema's `affected_area_sqm` field.

DeChant et al. [13] achieved 96.7% classification accuracy for Northern Leaf Blight on in-field maize imagery using a custom CNN, representing one of the strongest validations of CNN-based disease detection in true field conditions rather than controlled settings. Their work was limited, however, to a single disease and single crop type, with no cloud deployment infrastructure. The AgriDrone AI platform targets multi-crop, multi-disease generalization through its modular model architecture, where the YOLO leaf detector and disease classifier can be independently retrained and hot-swapped via the Python FastAPI model management layer without interrupting the frontend dashboard or database services.

The Retrieval-Augmented Generation (RAG) and knowledge graph paradigms from recent educational AI literature [18] offer an instructive architectural analogy for the AgriDrone AI pipeline: just as grounding LLM outputs in a structured knowledge graph improves factual accuracy in educational systems, grounding disease classification outputs in a structured knowledge base of confirmed disease signatures improves diagnostic reliability in agricultural systems. This principle motivates the AgriDrone AI architecture's separation of the raw inference layer (FastAPI + models) from the validated detection storage layer (Neon PostgreSQL `plant_detections` table), where detections progress through a confirmation workflow: detected → confirmed → treated → resolved.

Table 2. Comparative Analysis of Deep Learning Models for Plant Disease Detection

Reference	Model Used	Dataset	Accuracy / mAP	Inference Speed	Key Finding & Gap
AgriDrone AIAI(Proposed)	YOLO + PlantDiseaseCNN	Field-collected + PlantVillage	~94% (target)	2–5 sec (cloud)	3-model pipeline: detect → classify → boundary; cloud
Mohanty et al. [8]	AlexNet / GoogLeNet	PlantVillage (54K images)	99.35%	Offline batch	Benchmark CNN study; lab conditions only, field gap
Ferentinos et al. [9]	Custom CNN	PlantVillage variants	96.30%	Offline	Early drone CNN; no real-time inference or cloud
Deng et al. [2]	YOLOv3	Custom aerial dataset	88% mAP	Real-time	First real-time aerial YOLO pest detection; no disease
Themozhi & Rajesh [10]	MobileNet	Custom leaf images	90.20%	~3 sec (cloud REST)	Mobile-cloud REST inference; single model, no
Tm et al. [11]	CNN + SVM hybrid	Rice disease images	95.48%	Offline	High accuracy rice diseases; no generalization to other
Rangarajan et al. [12]	AlexNet + VGG16	Tomato disease dataset	91.20%	Offline	Transfer learning for tomato; no deployment architecture
DeChant et al. [13]	CNN (custom)	Maize field images	96.70%	Offline	In-field Northern Leaf Blight; no cloud or drone integration

3. Cloud Computing and IoT Architectures in Precision Agriculture

The integration of cloud computing with agricultural sensing systems has been surveyed extensively by Liakos et al. [14], who reviewed over 50 machine learning studies in precision agriculture and identified serverless database architectures and REST microservice patterns as the emerging standard for scalable agricultural data platforms. Their survey noted that the majority of deployed systems relied on relational databases with geospatial extensions for field boundary and detection coordinate storage, consistent with the AgriDrone AI AIAI choice of Neon PostgreSQL — a serverless, auto-scaling PostgreSQL variant that provides native JSON column support for the `boundary_coords`, `flight_path`, and `waypoints` fields in the AgriDrone AI data model.

Wolfert et al. [16] examined big data architectures for smart farming, highlighting the use of cloud IoT platforms such as AWS IoT Core and Azure IoT Hub for aggregating sensor streams from heterogeneous agricultural devices. Their analysis identified the absence of unified fleet management — the ability to coordinate, schedule, and monitor multiple autonomous devices from a single interface — as a critical gap in existing agricultural IoT deployments. The AgriDrone AI platform directly addresses this gap through its drone fleet management schema, which tracks each drone's status, `battery_level`, GPS coordinates, altitude, and home position in real-time, enabling multi-drone mission coordination from the Next.js dashboard's fleet view.

Themozhi and Rajesh [10] demonstrated a cloud-hosted REST inference architecture for plant disease detection, achieving 90.2% accuracy and 3-second end-to-end latency using MobileNet served via a REST API. Their work validates the cloud REST inference model adopted by AgriDrone AI but was limited to a single-model, single-endpoint architecture without fleet management, mission planning, or persistent geospatial storage. The AgriDrone AI API layer, built on Next.js Route Handlers, exposes eleven distinct endpoint groups covering drones, fields, flights, missions, alerts, detections, sensor data, plant detection, boundary detection, dashboard statistics, and AI inference, forming a comprehensive data orchestration layer between the drone hardware and the cloud database.

Kamilaris and Prenafeta-Boldú [15] reviewed 40 deep learning agriculture studies and observed that the transition from research prototype to production deployment was the single most frequently cited challenge, with data pipeline architecture, model serving latency, and hardware integration being the three primary barriers. The AgriDrone AI software stack directly confronts each of these barriers: the FastAPI Python backend with Uvicorn/Gunicorn ASGI serving addresses model serving latency; the Neon PostgreSQL serverless architecture addresses data pipeline scalability; and the open-source Pixhawk ecosystem addresses hardware integration transparency.

Barbedo [17] conducted a comprehensive survey of image processing methods for plant disease detection and specifically identified the absence of end-to-end deployment architectures — from image capture through AI inference to agronomic recommendation — as the most significant gap between academic research and practical field adoption. The AgriDrone AI platform closes this loop through its recommendations field in the plant_detections schema, where the AI inference pipeline populates actionable treatment recommendations alongside each disease detection record, accessible to farm operators through the dashboard detections page.

The serverless architecture pattern chosen for AgriDrone AI deserves specific discussion in the context of agricultural deployment economics. Traditional cloud deployments on reserved virtual machine instances incur fixed costs regardless of utilization, which is problematic for agricultural platforms exhibiting strong seasonal usage cycles — high activity during planting and disease-risk growing seasons, low activity post-harvest. Vercel's serverless Next.js deployment and Neon's serverless PostgreSQL both implement consumption-based billing, ensuring that infrastructure costs scale proportionally with actual usage. The Neon connection pooling layer handles the burst-reconnect patterns of serverless function invocations, which would otherwise exceed PostgreSQL's connection limits under concurrent drone data upload scenarios.

Table 3. Comparative Analysis of Cloud and System Architectures in Agricultural AI Platforms

System	Cloud Platform	Database	Frontend	API Layer	Architecture Characteristic
AgriDrone AI (Proposed)	Vercel (Serverless)	Neon PostgreSQL	Next.js 16 + React 19	FastAPI + Next.js Routes	Full cloud: fleet mgmt, AI inference, geospatial DB, real-time dashboard
Themozhi & Rajesh [10]	Custom REST Server	Not specified	Mobile App	REST API	Single-model cloud inference; no fleet or mission management
Liakos et al. [14]	Literature Survey	Varies	Varies	Varies	Surveys ML in agriculture; no unified platform proposed
Sa et al. [1]	Hybrid (ROS + Cloud)	Custom DB	ROS Visualizer	ROS Topics	Partial cloud; heavy onboard dependency, no web dashboard
Kamilaris & Prenafeta [15]	Survey	N/A	N/A	N/A	Reviews 40+ DL agriculture studies; no system deployment
Wolfert et al. [16]	IoT Cloud (AWS/Azure)	Time-series DB	Web Dashboard	IoT Gateway	Big data farm platform; sensor focus, no AI disease detection
Barbedo [17]	Not deployed	Image repository	None	None	Disease detection survey; emphasizes deployment gap in literature

PROBLEM STATEMENT

3.1 INTRODUCTION

Food security relies heavily on agriculture, but agriculture still suffers from problems related to inefficient management and monitoring. Current traditional methods of monitoring agriculture use manual inspection so that farmers physically check the health of plants, identify illnesses, and assess the conditions of their fields. Unfortunately, although the farmers have relied on this method for decades, it is often labor-intensive, takes a lot of time, and is inaccurate - especially for larger areas of land.

As demand increases for increased production on farms, the limits of relying on hands-on management are becoming more significant. Many times, farmers can't identify early signs of illness in an agricultural crop until the disease has spread beyond the point of no return, resulting in large production losses and greater use of chemicals. Declining production or losing the ability to use those chemicals can create economic hardship as well. Once crops have been destroyed, farmers have little chance of collecting debts for production or loss of use of those chemicals. In some cases, accurately establishing field boundaries and plot segmentation can be extremely challenging or impossible because of how the land is divided by natural barriers like rivers or hills.

However, there is hope for the agricultural community due to the advancement of new technologies, including AI, computer vision, and drones. With these technologies, the agricultural community will get an opportunity to transform how they manage their farms as well as their environmental impact on producing food locally and globally. High-resolution aerial images of agricultural fields can be taken quickly using drones with cameras. This aerial imagery of crop health, disease presence and field structure can help with large-scale data collection when combined with AI-based image analysis techniques.

There is currently no integrated solution that combines all three components (aerial data acquisition, intelligent analysis, and user-friendly visualization) into one system. Most of the solutions available today either only collect data, or offer limited analysis capabilities without being implemented in real-world agricultural applications.

Thus, there is a significant need for an automated, intelligent, scalable and cost effective solution that can leverage drone technology and AI to enhance agricultural monitoring. The solution must also automate data collection, provide accurate analyses and actionable insights through a user-friendly interface. AgriDrone AI will provide a solution to the challenges above by developing a drone-based AI agricultural monitoring system that will accomplish the detection of crops diseased and identify field boundaries.

3.2 BACKGROUND AND MOTIVATION

Over the past few years, artificial intelligence (AI) and remote sensing technologies have been used together successfully to improve agricultural productivity and efficiency. Advances such as deep learning, convolutional neural networks (CNNs), and object detection models (e.g. YOLO – you only look once) have provided unprecedented accuracy for image classification and detection tasks and will thus allow for improved accuracy when diagnosing crop diseases, classifying the conditions of plants, as well as analyzing field patterns.

Similar advancements in edge detection and image segmentation, such as the DexiNed edge-detection model (Dense Extreme Inception Network for Edge Detection), have led to improved precision when determining boundaries and other structures in images and will also enable more accurate identification of the field boundaries, the segmentation of fields, and estimation of land areas from aerial images for the purpose of facilitating efficient farm management and planning.

At the same time, technological advancements have led to a rapid evolution of drone technology, which has become cost-effective and easily accessible and has become a viable tool for obtaining high-resolution aerial imagery for large-scale agricultural applications. Drones in agriculture can fly across large areas in much less time than is needed for manual inspections, making them excellent tools for data collection in precision agriculture.

Despite advancements in technology, there remain some unresolved issues with current agricultural data solutions:

- Many of the existing systems that are available today have not fully integrated all their data together into one system, which requires farmers to use many different computer programs to collect, process and visualize their data.
- Often, there are both no real-time or close to real-time analysis challenges in current systems making them less useful in agricultural farm environments that may be constantly changing.
- Most ways of delivering solutions today do not provide a user-friendly interface so that farmers can easily use them in an environment where they need to be used.
- There are very few, if any, ways to provide low-cost solutions to small-scale and medium-scale farmers.

The goal of this project was to help bridge all of these existing gaps by creating one unified system for combining drone-based data acquisition with AI-based analysis and an interactive dashboard to create a communication channel to help farmers and other people involved in agriculture to have an effective way to monitor crop health, detect disease early and understand the structure of the field.

Additionally, the early detection of crop diseases is one of the most important tools to help farmers avoid the loss of crop yields and reduce the unnecessary use of chemical treatments because they will be able to utilize targeted pesticide applications as opposed to broad applications made over the entire field. This will lead to more sustainable practices in agriculture.

One of the main motivators behind accurate identification of property lines is to better understand how land is divided into smaller plots as well as measuring the respective area of each plot for better management of resources, irrigation and estimating potential yield. Presently, the methods used in measuring land are typically inefficient and not very accurate; even more importantly when dealing with larger plots of land or oddly shaped plots.

Along with the benefits of accurate identification of property boundaries there is also the fact that there is a growing demand for agricultural products to be produced digitally; more specific to these trusted brands, there is an increase in the need for utilities that facilitate these activities such as precision farming and smart agriculture. Worldwide the government and organizations are working together to increase the use of technology in farming to help increase the quality of the product while increasing the quantity of the product produced for sells; this system was designed to assist in respecting these goals by being an easy and effective, yet cost effective way of using AI, Digital Technology, as well as drones within our daily farming tasks.

3.3 PROBLEM DEFINITION

Agricultural monitoring practices are manual and inefficient, do not provide a high degree of accuracy in their results and do not scale to meet the demands of today's farms. Farmers frequently visually inspect their crops for signs of disease and to assess the condition of their fields; this is a time-consuming and error-prone method of assessing both the overall health of a crop/field and its specific needs as the area of land being farmed continues to grow larger.

While many of the existing technological solutions show promise; these solutions suffer from limitations that greatly reduce their effectiveness in reality. Many of the current agricultural monitoring systems are focused on specific types of functions (e.g. disease monitoring, aerial imaging) and therefore do not offer farmers a full suite of solutions that are integrated. An additional issue with many of today's agricultural monitoring solutions is that they do not work reliably in the field (i.e. when measuring accurately in the presence of variations in the quality of light, type of crop, or environmental conditions).

One large reason for the delay in accurately detecting an agricultural disease is that most agricultural monitoring systems require the camera to take close-up images of the affected crop; however, many systems do not have the capability to detect subtle disease patterns that may develop in early stages of an outbreak. Without timely detection of crop diseases through agricultural monitoring, producers risk losing a significant portion of their crop to disease, while continuing to rely on the use of synthetic pesticides to control crops' pests.

Field boundary detection is still an area where farming remains difficult. Using traditional methods to measure land can be time-consuming, and these methods will not provide a clear separation of the plots. In addition, with boundaries that lack definition, it can be even more difficult to separate fields. Current automated solutions can struggle with complicated field structures, overlapping crops, or complex image contrast between fields and crop types.

There also aren't any easy-to-use interfaces for users to visualize or analyze their raw agricultural data. When users don't have the right platform, even accurate data will not help them make decisions.

In light of these limitations, this project will solve four major problems:

- The need for a highly efficient and scalable monitoring method.
- The need for a method of accurately and quickly detecting crop diseases through aerial imagery.
- The need for accurate field boundary detection and plot segmentation.
- The need for integrated platforms for data acquisition, processing, and visualization that work well.
- The need for a cost-effective, user-friendly solution for use in agriculture.

This project has developed an AI-based drone platform for precision agriculture that combines different functions into one solution. The platform will use computer vision to identify and segment crop disease and will feature an online dashboard for data visualization and analysis.

With the new system, users will gain real-time access to accurate and efficient agricultural monitoring data. By combining drone technology and artificial intelligence, we will help farmers adapt their traditional farming methods to more data-driven and intelligent practices.

In the end, our proposal addresses the limitations of current biodiversity assessing methods--and helps advance the field of precision agriculture by providing farmers with actionable insights that support informed decision-making; reduce crop loss; and ultimately increase productivity.

PROPOSED SOLUTION

4.1 SYSTEM ARCHITECTURE

- **Drone Hardware System**

A hardware system is important as it is the foundation of AgriDrone AI. The drone uses an S500 frame, 935KV brushless motors, 1045 propellers, 35A ESC's and Pixhawk 2.4.8 flight controller to provide stability, control and manoeuvrability for flight. The M8N GPS module provides location to track, the T10 transmitter and R10 receiver provide manual control of the drone, and the camera on board allows for remote capture of image data to feed into one AI.

- **Image Acquisition Process**

The image capture portion of the system begins when the drone is used to acquire images of agriculture using the camera and is flown using manual controls by the operator. Images must be acquired from multiple positions and ranges and the system is designed to allow for efficient coverage of large areas with uploaded images after collection. Image quality is a critical aspect of the AI predictive modelling since image accuracy is based on lighting, altitude and the angle of the camera (to the earth).

- **Input handling and image transmission**

Images are transferred to the system via the web interface after they have been captured. Uploading images using the graphical user interface is simple; through API calls, images are transferred from the front end to the back end. The system manages how image data is handled correctly, including format validation and preparing the data for processing by AI models.

- **Back end processing framework**

FastAPI is the framework used to create the back end of the system. It serves as the main processing unit (the back end) of the system, where image data is received from the front end, routed to the proper AI pipelines, and ultimately returned as answers after model loading and AI inference processing. The back end manages how different parts of the application communicate with one another and how each part of the application can handle many separate requests yet achieve high levels of performance.

- **The Plant Disease Detection module**

It is designed to assess crops for signs of disease. Using a YOLO-based model, this module identifies and isolates the leaf areas from the images; once isolated, additional processing is performed using RGB or HSV filtering and contrast enhancement. After the processing, the image will be sent to a convolutional neural network that can classify the plant as either healthy or diseased, as well as return with a corresponding confidence level regarding its determination. The result of this analysis is then passed back to the frontend for display purposes.

- **The Field Boundary Detection module**

It provides an automated way to detect the field watershed boundaries and separate the farmland into individual plots based upon these boundaries. The Dexin Curvature Edge Model is used to define these field boundaries from aerial images, and segmentation algorithms are then applied to separate the plot into smaller pieces. During this process, the area for each plot will also be calculated, which can help farmers manage their fields and plan for use of their land.

- **Database & Data Storage**

The system consists of a PostgreSQL database where all the required information will be held, including Detection results, Field data, and System records. Users will be able to view historical data and keep track of the data for a given period of time, providing them with a systematic method of managing their agricultural information. The database will also support organized storage of data and make it easier to retrieve data when needed quickly.

- **Frontend & User Interface**

The Front End will be created using Next.js and React, with an interactive and user-friendly interface that enables Users to upload Images, run detection processes, and visualize their results in a structured format. The visual results of the analysis will include Segmented Maps of Fields and results of disease detection, making it easier for the User to understand and comprehend the analysis. Other important metrics, such as Confidence Scores of detection, Number of Plots analyzed, and Area sizes, will be available on the User's interface.

- **Workflow & System Integration**

The entire system is designed to work together and be completely integrated. The Drone captures an image of the field, then the User uploads it to the Front End. The Back End receives the image and processes it using AI Detection Models, and the results of the Detection are stored in the database and sent back to the Front End for visualization. The overall workflow of the system is completely seamless, resulting in efficient Data Processing and projection of crop health & field structure to the User in a timely and accurate manner.

4.2 HARDWARE SYSTEM DESIGN (DRONE COMPONENTS)

- **Drone Frame (S500 Frame)**

The S500 frame serves as the structural foundation of the drone, providing support for all hardware components. It is lightweight, durable, and designed to maintain stability during flight. The frame ensures proper placement of motors, flight controller, battery, and other modules, allowing balanced weight distribution and efficient aerodynamics.

- **Brushless Motors (935KV)**

The drone uses 935KV brushless motors to generate the required thrust for flight. These motors convert electrical energy into mechanical rotation, spinning the propellers at high speed. The KV rating determines the motor's RPM per volt, ensuring optimal performance for lifting the drone and maintaining stable flight conditions.

- **Electronic Speed Controllers (35A ESC)**

Each motor is connected to a 35A Electronic Speed Controller (ESC), which regulates the speed of the motor based on signals received from the flight controller. The ESC plays a crucial role in controlling thrust, enabling precise adjustments in speed and direction for stable flight.

- **Flight Controller (Pixhawk 2.4.8)**

The Pixhawk 2.4.8 acts as the brain of the drone. It processes data from onboard sensors such as gyroscopes and accelerometers to maintain stability and control orientation. It also manages communication between different components and ensures smooth flight operation by adjusting motor speeds in real time.

- **GPS Module (M8N GPS)**

The M8N GPS module provides real-time positioning data, including latitude, longitude, and altitude. This enables location tracking of the drone during flight and supports mapping and analysis tasks. GPS data can also be used for future extensions such as autonomous navigation and waypoint-based missions.

- **Transmitter and Receiver (T10 & R10)**

The T10 transmitter and R10 receiver form the communication link between the user and the drone. The transmitter allows the user to control the drone's movement, including direction, altitude, and speed, while the receiver on the drone interprets these signals and passes them to the flight controller for execution.

- **Power Supply System (LiPo Battery – 4S 5200 mAh)**

The drone is powered by a 4S 5200 mAh Lithium Polymer (LiPo) battery, which provides the necessary electrical energy for all components, including motors, flight controller, and GPS module. The battery ensures sufficient flight time and stable power delivery during operation.

- **Power Management Module (PM02)**

The PM02 power module regulates and distributes power from the battery to different components of the drone. It also monitors voltage and current levels, helping prevent over-discharge and ensuring safe operation of the system.

- **Camera Module (Data Acquisition Unit)**

A camera is mounted on the drone to capture high-resolution aerial images of agricultural fields. This camera acts as the primary data acquisition unit, providing input for the AI-based analysis. The quality of captured images directly impacts the accuracy of disease detection and boundary segmentation.

- **Battery Charging and Safety Components**

The system includes supporting components such as an iMAX B6 charger and a voltage tester to ensure safe battery charging and monitoring. These components help maintain battery health, prevent overcharging, and ensure reliable operation of the drone during field use.

Overall, the hardware system is designed to provide stable flight, efficient data collection, and reliable performance. Each component plays a specific role, and their integration ensures that the drone can effectively capture high-quality images required for intelligent agricultural analysis.

4.3 DATA COLLECTION AND IMAGE ACQUISITION

- **Aerial Imagery - The Process of Capturing Aerial Imagery Using Drones**

In order to collect data through aerial imagery, drones are flown over the farming areas to create a series of aerial images, this is a method that allows for covering large areas quickly and easily while being versatile in how the drone can navigate, the quality and coverage of the images will greatly impact how accurately the AI models will be able to analyze them.

- **Data Collection Using Cameras Mounted on Drones**

The cameras mounted on the drones will be used to acquire high resolution images of crops and structures in the field—these images contain a significant amount of detailed visual information, the images will include leaf texture and variations in color between leaves, and the boundaries of fields. The images collected by the drones will be used as the main source of input into the processes of detecting plant diseases and segmenting the field boundaries. Positioning the camera correctly and maintaining stability will help eliminate motion blur, which is essential to ensure that the images are clear.

- **Image Capture Conditions**

In order to ensure that the images captured will provide reliable results, the images need to be captured in conditions such that the environmental factors - such as light, weather, altitude and camera angle—will produce good images. Generally, images will be captured in daylight hours when little or no shadows and moderate wind exist in order to ensure that they are clear and that they have used the same altitude as other images, in order to maintain the same scale for all of the images, this is critical when it is important to detect the field boundaries accurately.

- **The Overlapping and Coverage Strategy**

Data collection using the drone involved capturing multiple photographs with small amounts of overlap. Overlap assists in covering all areas of the field, decreasing potential for missing sections and increasing the robustness of boundary identification.

Also, overlap aids in providing more uniform visual information over both adjacent areas and helps provide more effective segmentation.

- **Data Transmission to System**

Photos taken by the drone will be transmitted to the system once they are acquired, either by physically retrieving them from the camera or through uploading them via the web interface. The front end of the interface has an easy-to-use upload mechanism, enabling users to quickly select and upload images for processing.

- **Image Validation and Preprocessing Resolution**

The system will also perform basic validation on the photos taken by the drone after images are uploaded. Basic validation entails confirming that files have valid formats, dimensions and resolution before sending them for further processing. Basic validation ensures that input data meets requirements for the AI models being used, which in turn binaural avoids processing errors.

- **Collection and Preparation of a Dataset for Analysis**

The collected photographs have been placed into a well-organized dataset which can be used for future reference and analytical purposes. Properly organizing the photographs allows for rapid processing of data, easy access and greater overall management of agricultural datasets. This dataset can also be used for enhancing and retraining machine learning models as needed.

- **The Importance of Data Quality**

The overall success and operation of the entire AgriDrone AI system will depend heavily upon the quality of the photos that have been collected. Higher quality (resolution, lighting) and more accurately framed images will produce the best possible results in relation to disease detection and boundary segmenting of the crop area being monitored. Conversely, lower-quality images will result in poorer quality predictions and decrease the reliability of the overall AgriDrone AI system.

Overall, the way that data and images are collected and assembled into a usable dataset will form a strong foundation for the success of the AgriDrone AI system. By providing high-quality and accurately formatted data for use in the system, the AgriDrone AI will be able to provide accurate and useful information to aid in monitoring agricultural activities and making informed business decisions about those activities.

4.4 PLANT DISEASE DETECTION PIPELINE

- **Leaf Recognition via YOLO**

The first step of the pipeline will be the identification of leaf areas within the input image using a YOLO (You Only Look Once) Object Detection Model. By using the provided bounding boxes for leaf areas, the model isolates all other regions of interest that may appear in the input image (e.g., dirt, clouds, etc.) to focus solely on the leaf areas present.

- **HSV Validation**

After recognizing the leaf areas, the next step will include applying HSV (Hue, Saturation and Value) Color Filtering to confirm that the identified leaf areas are indeed vegetation. This helps to remove non-plant items from future analysis to create an even more accurate measurement for detecting the presence of leaf areas.

- **Preprocessing/Enhancement of Image Quality**

The identified leaf areas will also be preprocessed to assist with improving Image Quality. Many techniques can be used, but CLAHE (Contrast Limited Adaptive Histogram Equalization) may be the best suited for enhancing the visibility and contrast of leaf areas, since lighting can vary greatly from day to day.

- **Feature Extraction**

Once the image has been processed, the next step will be to determine all of the relevant visual characteristics of the processed image (e.g., color variations, structural differences, texture, etc.) that will help to differentiate a healthy plant from a diseased plant.

- **Classification Using CNN**

The features that were extracted from the plant image are then input into a convolutional neural network (CNN) that has been trained on multiple datasets of plant diseases. The CNN is able to evaluate the features and classify whether the plant is in a healthy or diseased condition. In addition to providing classification output, the CNN also provides a confidence score associated with the prediction made by the model.

- **Decision Logic**

The system utilizes the confidence threshold to arrive at the final output of the application. If the confidence score for the prediction is greater than a predetermined threshold, the plant is classified as being diseased; otherwise, the plant is classified as healthy. This guarantees a reliable and consistent output of predictions.

- **Set of Outputs**

The output of the CNN model will include the classification output (either healthy or diseased), the confidence score associated with the classification output, and the bounding box of the leaf region detected by the CNN algorithm. This output is returned to the front-end application and displayed to the user in a user-friendly manner.

- **Storing and Tracking Results**

The results of the detection of the plant image can be saved in the database for future reference and analysis. By saving this information, user will be able to monitor their plants over time and identify trends or issues that continuously occur within a particular area.

Overall, the plant disease detection pipeline enables automated and accurate analysis of crop health by combining object detection, image processing, and deep learning techniques. It plays a crucial role in early disease identification, helping farmers take timely action and reduce potential crop losses.

4.5 FIELD BOUNDARY DETECTION AND SEGMENTATION

- **Aerial Imagery Input**

Farm land area aerial photographs have been taken with drones; or uploaded via an Internet user interface. These photographs contain many separate agricultural areas containing visible/ semi-visible edges i.e. paths, lines of soil, differences in crops.

- **Image Pre-processing**

Image pre-treatment takes place before applying the border edge detection process. The image was subject to pre-treatment to create improved images. Pre-processing may include resizing, reducing noise and improving the contrast of the image. This means an individual image is subjected to pre-processing makes the edges of important items and structural characteristics of that image easier to distinguish to increase the accuracy of border edge detection.

- **Edge Detection using DexiNed**

The boundary edge detection process is accomplished using DexiNed (Dense Extreme Inception Network for Edge Detection). DexiNed is a new state of the art Deep Learning model which can accurately identify and delineate complex and fine edgemarks in an image. DexiNed identifies boundaries in images of separated areas by identifying edges, these edges would be identified as the points between individual crops, points on irrigated lines, and the edge where there is an output difference in image brightness from soil to crop.

- **Boundary Extraction and Segmentation**

The edges detected by the model are refined by removing weak or irrelevant edges while keeping strong and continuous boundaries. These refined edges are then used by segmentation algorithms to divide the entire field into clearly defined and separate plots, making the structure of the land easier to analyze.

- **Labeling, Visualization and Area Calculation**

Each segmented plot is assigned a unique label and displayed using different colors for clear visual distinction. The system then calculates the area of each plot by converting pixel values into real-world measurements, providing accurate information useful for planning and management.

- **Result Output and Data Storage**

The final results include the total number of plots, individual plot areas, and a segmented image showing clear boundaries. These results are displayed on the dashboard in an organized manner and are also stored in the database, allowing users to track changes over time and perform further analysis.

4.6 BACKEND PROCESSING AND API INTEGRATION

- **Backend Functionality with API Communication**

FastAPI is used as the primary structure for the back end of the system, which serves as the main processing unit for the system. All front-end NEXUS to process requests from users to back end are done using API endpoints—this allows for seamless communication between user interface (UI) of the app and back end of the app processing. Every time the user selects an image or initiates analysis, the request will then go through the API path to the back end.

- **Incoming Request & AI Model Handling**

When an incoming request arrives at the back end it will determine which internal processing module should handle it. For example, if the request is for a plant disease detection or for detecting the edge of a boundary. All requests will be processed using an AI (e.g. YOLO, CNN, DexiNed) model that will provide model performance information back to the back end for the final processing of the image for the prediction result. In addition, all model processing will be executed and inference will be made on the input images in order to provide accurate prediction results.

- **Data Processing and Response Generation**

After processing, the backend converts the output into structured data, including classification results, confidence scores, and plot details. These results are sent back to the frontend in formats such as JSON, allowing them to be easily displayed on the dashboard.

- **Database Interaction and System Integration**

The backend communicates with the PostgreSQL database to store and retrieve data such as detection results and field information. It also handles validation and error management to ensure reliable operation. Overall, the backend acts as a bridge connecting the frontend, AI models, and database, enabling smooth and efficient system functionality.

4.7 FRONTEND SYSTEM AND USER INTERFACE

- The frontend is developed using Next.js and React to provide a fast, responsive, and scalable user interface for interacting with the system
- Tailwind CSS is used for styling, ensuring a clean layout, consistent design, and improved user experience across different devices
- The interface includes multiple modules such as dashboard, live disease detection, boundary detection, and analytics for organized navigation
- Users can upload aerial images through an intuitive input system, which sends data to the backend via API calls for processing
- Real-time visualization is provided for results, including disease detection outputs with confidence scores and segmented field maps with labeled plots
- Interactive components such as charts and maps are used to represent data clearly, helping users understand crop health and field structure
- The system displays key metrics such as number of plots, area measurements, and detection results in a structured dashboard format
- The frontend dynamically updates based on backend responses, ensuring smooth data flow and immediate result display
- Error handling and validation are implemented to guide users during image upload and processing steps
- Overall, the user interface is designed to be simple, interactive, and efficient, allowing users to easily analyze agricultural data without requiring technical expertise.

4.8 DATABASE DESIGN AND DATA MANAGEMENT

- The system uses a PostgreSQL database to store and manage all application data in a structured manner
- The database includes entities such as drones, fields, detections, and system records to organize agricultural information efficiently
- Detection results such as plant health status, confidence scores, and segmented plot data are stored for future reference
- Field-related data including boundaries, plot areas, and crop information are maintained to support analysis and tracking
- The database enables historical data storage, allowing users to monitor changes in crop health and field structure over time
- Efficient querying and indexing techniques are used to ensure fast data retrieval and smooth system performance
- The backend communicates with the database to insert, update, and retrieve data as required
- Proper data management ensures consistency, reliability, and scalability of the system for future enhancements

4.9 SYSTEM WORKFLOW AND IMPLEMENTATION

- The system workflow begins with the drone capturing aerial images of agricultural fields during manual flight
- Captured images are uploaded to the frontend interface by the user for analysis
- The frontend sends the images to the backend through API requests for processing
- The backend routes the data to appropriate AI pipelines, including plant disease detection and boundary detection modules
- The disease detection pipeline analyzes leaf regions and classifies them as healthy or diseased with confidence scores
- The boundary detection pipeline identifies field boundaries, segments plots, and calculates their respective areas
- Processed results are sent back to the frontend in a structured format for visualization
- The frontend displays outputs such as segmented field maps, plot details, and disease detection results
- Relevant data is stored in the database for future access and analysis
- The complete implementation ensures seamless integration of hardware, software, and AI components, enabling efficient agricultural monitoring and decision-making

EXPERIMENTAL SETUP AND RESULT ANALYSIS

5.1 OBJECTIVE

The goal of the experimental setup is to evaluate the performance of the AgriDrone AI system for intelligent agricultural monitoring. The AgriDrone AI system captures drone aerial images, analyzes them using AI, and provides valuable insights into crop health and structure.

This experiment is designed to validate two functional characteristics within the AgriDrone AI system:

- Accurate identification of crop disease using AI models
- Consistent identification and segmentation of agricultural fields.

The secondary goal of this experiment is to evaluate the processing performance, usability, and reliability of the AgriDrone AI system in a real world agricultural management environment.

5.2 AIM

This experiment's purpose is to design and evaluate a comprehensive system that utilizes AI-driven drones to collect data and use computer vision to analyze through integrated drones, collect data, analyze the data using computer vision from the drone-based images.

This system is intended to:

1. Automate the monitoring of crops from an aerial perspective.
2. Allow for the detection of plant disease in its early stages.
3. Accurately define each agricultural field as a function of the unique plot that it encompasses.
4. Conduct analysis and display results to users via an interactive dashboard.

The overall purpose of this project will be demonstrated using this combined AI-drones approach to demonstrate improved efficiencies and improved decision-making capabilities in agriculture.

5.3 SOFTWARE REQUIREMENT

Building the proposed solution involves combining machine learning frameworks, backend technologies, and frontend tools together.

Key software requirements include:

- Programming Language: Python
- Backend Framework: FastAPI
- Frontend Framework: Next.js
- Database: PostgreSQL
- Deep Learning Libraries: PyTorch
- Computer Vision Libraries: OpenCV, NumPy
- Visualization Tools: Matplotlib, custom UI dashboard

Each tool helps develop models, integrate APIs, store data, and visualize data.

5.4 PROGRAMMING ENVIRONMENT (PYTHON)

Python has a wide range of features that support machine learning, computer vision, and backend development, it is the primary programming language used for the development of the proposed system. The flexibility and abundance of libraries available to Python developers provide an environment for quick iteration through development and trial and error process.

The following environment will be created specifically for your development needs:

- Training & Inference with Models (in Python)
- API Integration with FastAPI
- Data Preprocessing and Image Handling Model Deployment either Real-Time or Batch

The modular structure of the Python language provides an easy way to integrate the different components of the system

5.5 IMAGE PROCESSING & DATA ACQUISITION

For the analysis of fields, the information is derived from drone-captured aerial images, with the drone being operated manually to get high-quality photographs of various portions of the field.

The following processing steps are applied to the images before passing through an AI model:

1. Acquisition - drone capture or upload point
2. Preprocess - resizing, normalization, and noise removal
3. Further enhance features extracted

These three steps above ensure that the data provided to the AI model for processing is of a sufficient quality to produce an accurate result.

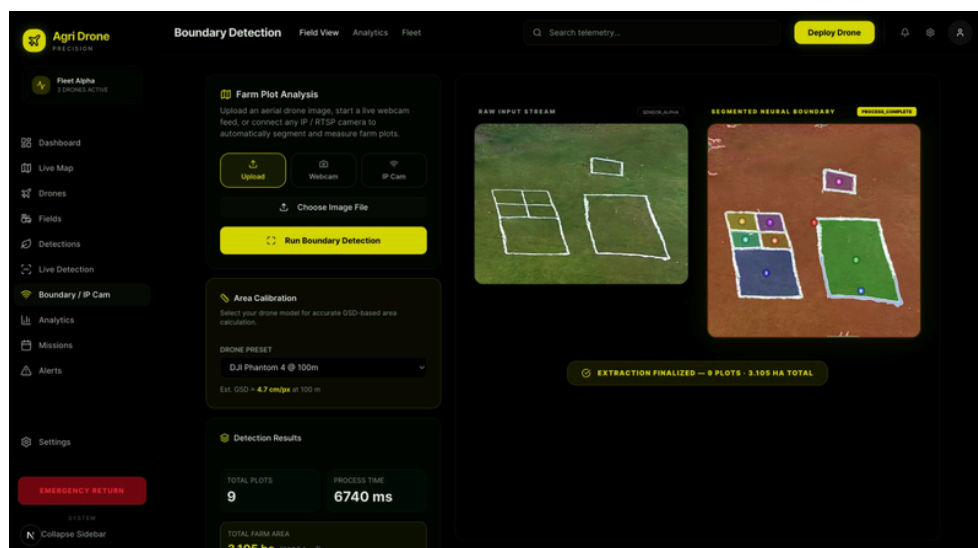


Figure 5.1: Image Processing and Data Acquisition Workflow

Aerial images are obtained via drone shooting or retrieving from other sources; in this case, the data will be pre-treated (resized, normalized, and noise-reduced) before being processed with a computer using AI technology to produce accurate results.

5.6 DISEASE DETECTION MODEL (YOLO + CNN)

The disease detection pipeline comprises two types of models, object detection and classification. YOLO (You Only Look Once) detects and locates the area of interest, e.g. a leaf or infected area.

A Convolutional Neural Network (CNN) classifies the detected area of interest as either healthy or infected. This gives a higher degree of accuracy in the detection of disease by combining detection and classification.

Key features of this method include:

1. bounding box detection for the infected region
2. confidence scores on predictions
3. classification of disease type or healthy class.

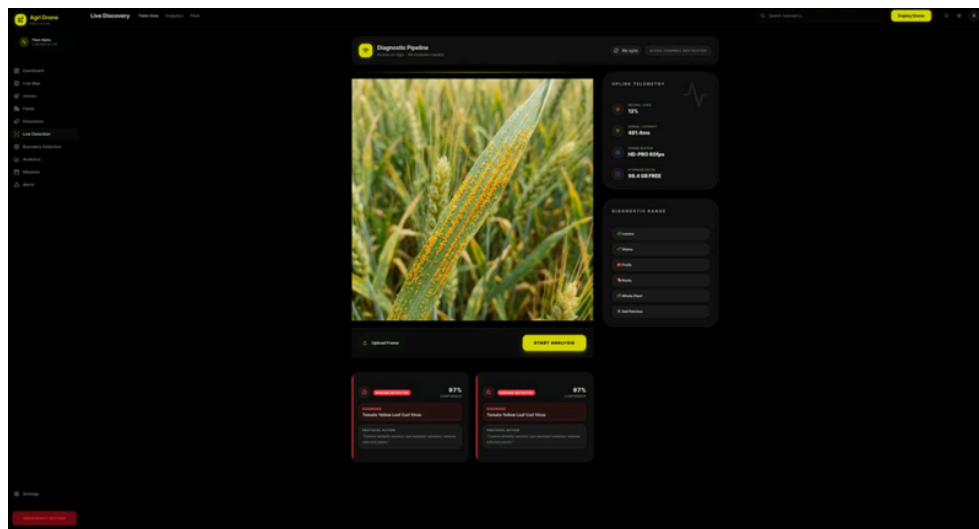


Figure 5.2: Disease Detection Interface Showing Infected Crop Analysis

The AgriDrone AI System's disease detection module is shown in Figure 5.2. This module uses an aerial or close-up photo of a crop leaf to diagnose any possible disease on that leaf. An AI pipeline takes the image uploaded by the user and passes it through a trained model, which outputs sick parts of the crop and overall condition of the crop.

The diagnosed image will be the most apparent part of the user interface, and other pieces of diagnosis information (e.g., confidence levels that the disease has been detected and a classification of the crop's condition) will be included in the interface for the user. When a confidence score ($> 80\%$) is shown, it tells the user how likely it is that the disease was correctly identified when combined with visual cues, such as colored labels and warning sections, showing the difference between diseased and healthy output.

The system will also have a summary of the problems provided below the image with severity levels. The data for the problems detected will be produced using a YOLO-based detection model and a CNN classifier to ensure that crop diseases are detected and accurately identified in both locations and classifications. As shown in this interface, the AgriDrone AI System can give users fast, understandable feedback on detecting diseases before they can spread. This will aid the user in intervening with pesticides as soon as possible.

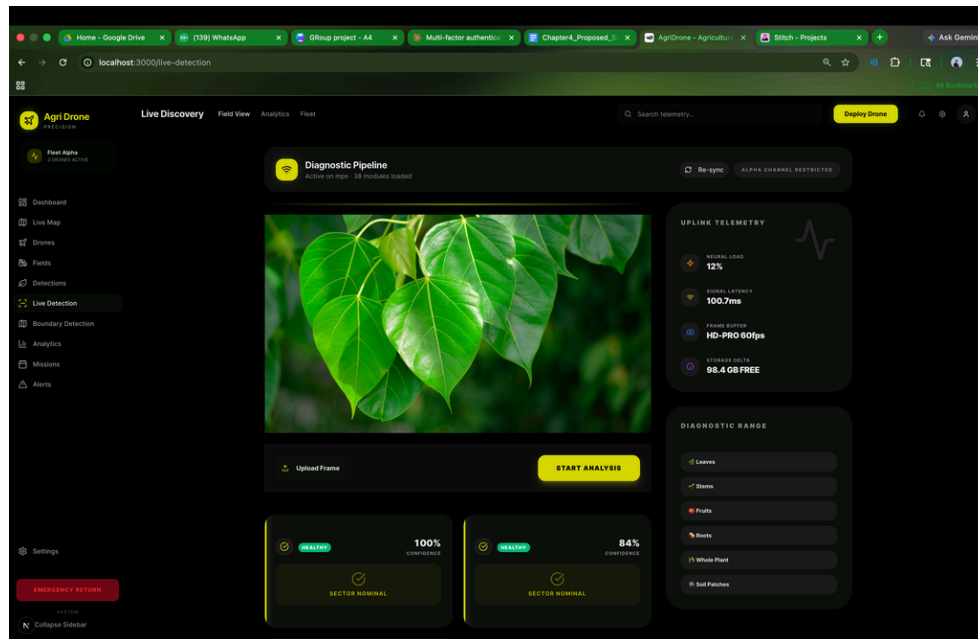


Figure 5.3: Diagnostic Pipeline Interface Showing Healthy Crop Classification

The diagnostic pipeline of AgriDrone AI's crop health assessment system can be seen in figure 5.3. This is an interactive dashboard-style interface comprising areas of the interface that allow users to interact with each part of the diagnostic procedure or tools. An example of an uploaded image in this interface shows a sample of healthy crops, and the AgriDrone AI system was able to correctly classify this sample as "healthy" with high confidence (e.g., 100% confidence).

The Diagnostic interface contains multiple sections (diagnostic area) including the image display section, the telemetry section, and the summary of diagnostics area, respectively. The telemetry section of the interface provides information about the data used by the AgriDrone AI system to make the classification of crop health, which includes processing data and overall calibration (status) of the system. The summary of diagnostics area contains classifications that are unique to a specific type of crop or crop disease and potentially lists options for diagnosing an incorrect classification (i.e., whether a crop is diseased).

The output section from the AgriDrone AI system lists the classification result and corresponding confidence level percentages, which indicate how reliable the classification was in determining whether the analysed crops were healthy or diseased. The clean presentation of elements in the output section was designed to enhance user experience and will provide an efficient means for the users to interpret their outcomes.

The clarity of this figure demonstrates how well the AgriDrone AI system can identify healthy crops, which in turn reduces false positive classifications and builds user trust in the classification results from AgriDrone AI.

5.7 BOUNDARY DETECTION (DEXINED + WATERSHED SEGMENTATION)

Detecting field boundaries is achieved by applying various advanced edge detection and segmentation techniques.

1. Using DexiNed (deep extreme inception network), we extract fine edges from aerial imagery.
2. Then we apply the Watershed algorithm for segmenting into distinct fields (plots).

The combination of these methods allows us to accurately and precisely identify field boundaries, even in complex or irregular layouts.

The outputs will be:

1. Segmented plots
2. Number of plots
3. Area of each plot

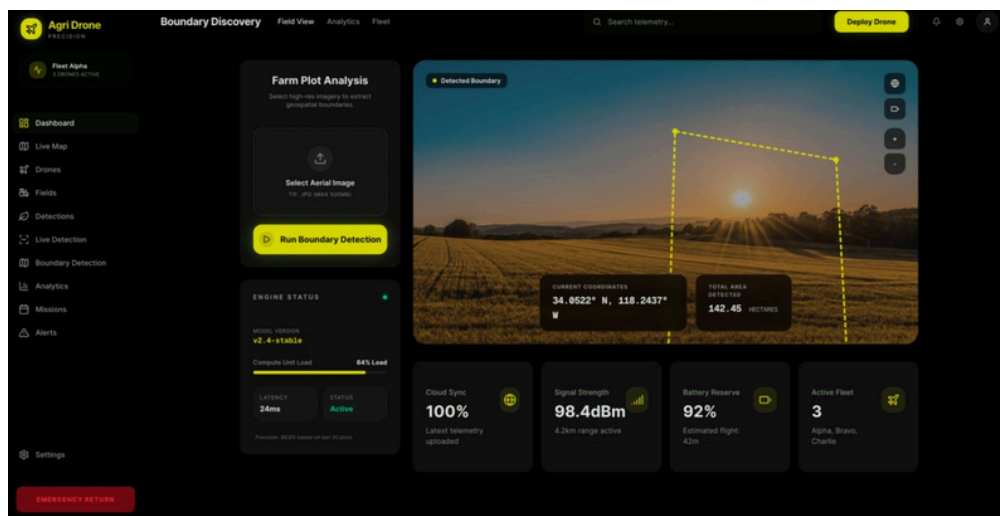


Figure 5.4: Boundary Detection Interface Showing Farm Plot Selection and Area Estimation

Figure 5.4 shows the agrarian boundary detection module of AgriDrone's machine-intelligence system for agronomy in which an agricultural field has been selected and analyzed to detect its boundaries and analyse its characteristics within the specified region. Aerial photographic images of the field with boundaries overlaid as polygons are displayed on the graphical user interface (GUI).

The user selects a designated area of interest from the GUI, and by selecting boundary detection; the system concludes the edges of the field and outlines the entire field with dashed or highlighted lines reflecting the designated boundary or edges of the field. The user can also see the geographical coordinates and perceived area measurements along with the image, which provides quantitative information about the field area selected in the application.

Other parameters of the AgriDrone AI system, such as signal strength, battery life, and altitude, are displayed in conjunction with the image, and are generated using real-time telemetry data collected from drone images. This integration of spatial analysis and operational data greatly increases the system's practical value. The results confirm that this system accurately identifies field boundaries and estimates field areas, which is vital for land management, resource distribution, and planning.

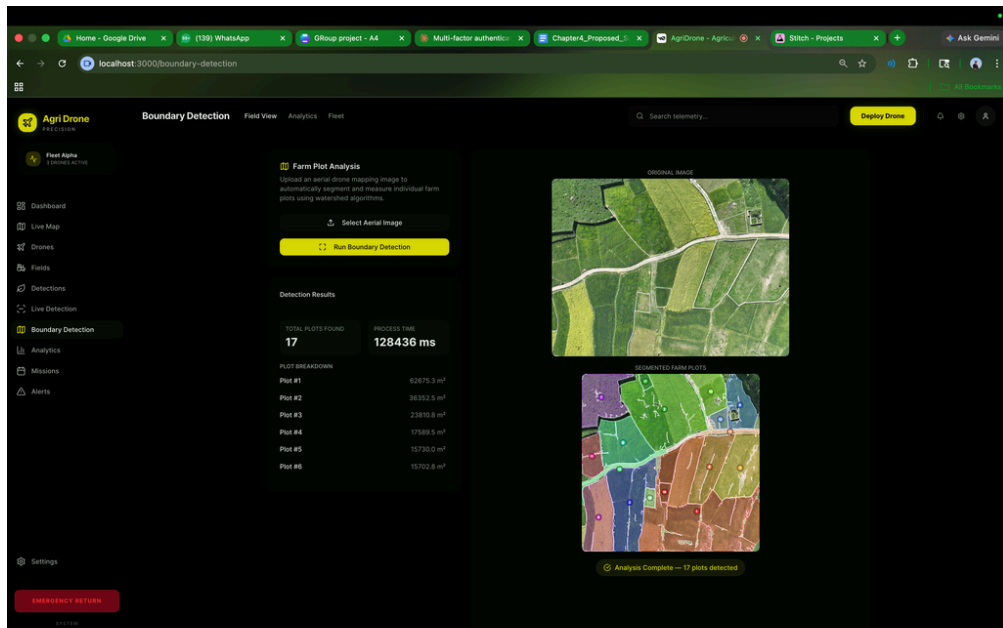


Figure 5.5: Boundary Segmentation Output Showing Multi-Plot Detection and Area Distribution

Figure 5.5 shows an example from an aerial shot that has gone through the process of enhancing the image and then segmenting it into separate parcels of land. The top half of the image shows the original image you would take of a field/land area in an agricultural area prior to segmentation, while the bottom half of the image shows how the original image has been segmented into several visually distinct images (due to various colors).

The system can detect several verifiable parcels of land and assign them to an individual space in the segmented images. Among other potential outputs, the system generates the total number of parcels identified in the original image, as well as the time it took to segment the original image and the spatial area of each segment. The above outputs can be calculated through the use of edge detection and segmentation processes, which allow the system to accurately identify the boundaries of parcels of land even when the edges of the parcels cannot be seen easily.

The results of the segmentation of the image show that the proposed method for segmenting adjacent parcels will be successful even when clear boundaries are not provided. This will provide accurate information about the location and boundaries of parcels in areas where ownership has been fragmented or land is farmed very close together (i.e. parcels are planted very close together).

The results of this image demonstrate how the proposed method will enable segment analysis of each parcel as a separate area, providing valuable information for agricultural planning and precision agriculture.

5.8 PERFORMANCE EVALUATION (PROCESSING TIME & ACCURACY)

We evaluate the performance of the system against key metrics on the efficiency and effectiveness of the results.

The evaluation parameters will include:

1. Time taken to process each image (Inference time)
2. The accuracy of disease classifications (Confidence of detection)
3. Clarity and correctness of boundary detection (Quality of segmentation)
4. Speed with which the dashboard is updated (Responsiveness of the system)

As well as optimising performance as we work under limited computational resources (free tier cloud services).

5.9 RESULT ANALYSIS

The test results show that the method proposed in this work has effectively carried out disease detection and boundary segmentation tasks.

For detecting diseases, the model provided an accurate identification of the condition of the crops; any crops were identified as either healthy or diseased with high confidence (e.g., between 80% and 99% confidence).

The boundary detection module accurately segments field images into multiple plots even for moderately-complex images.

The processing time for an image is a matter of seconds; therefore, it would work well in practical usage scenarios.

There were, however, some limitations to the results of our tests:

- Low-quality or blurred images will reduce our accuracy,
- Complex field structures will affect the accuracy of boundary segmentation,
- The processing speed will depend on the resources available to the processing unit.

The results of these tests confirm that the system is a reliable tool that can assist in monitoring within the agricultural community.

5.10 SYSTEM OUTPUT

Users receive output from a simple-to-use web dashboard that provides visual and numerical representations of results.

The features of the output include:

- Detection of disease with a bounding box and score for confidence
- Labels to classify as healthy or diseased
- Segmented maps of fields that are color coded
- Number of plots along with their distribution over an area
- Processing time along with log history

The dashboard will help the users interpret the results easily and will allow them to understand how to make decisions based on the results.

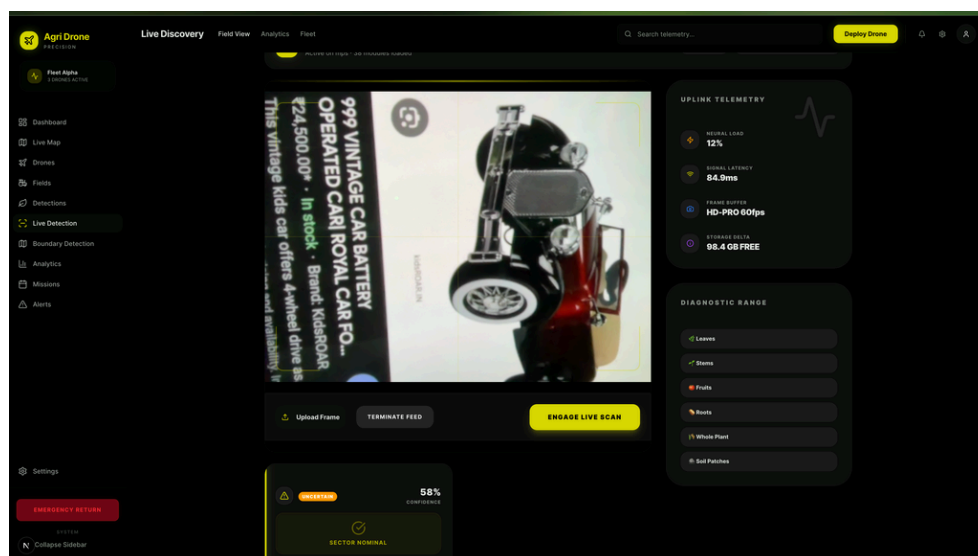


Figure 5.6: Diagnostic Interface Demonstrating System Robustness with Non-Agricultural Input

The image that is shown in Fig. 5.6 shows how the AgriDrone AI system diagnoses itself when given an image of an object that has no relation to crop data. This is a basis for how well the AgriDrone AI system works with images that are not agricultural and how it will respond to images that do not fit into the agricultural data category.

As a non-agricultural input image is processed using the exact pipeline that is used for the AgriDrone AI's agriculture analysis (including preprocessing, feature extraction, and classification) the output will show a low or uncertain confidence score on the detection of the agricultural crops or diseases from the input image. By having an uncertain confidence score on non-agricultural input images, the AgriDrone AI system does not give false confidence in classifying arbitrary objects as being valid agricultural data.

Included in the interface will be telemetry data that shows the signal strength, pixel count, and processing metrics. This information will add transparency in the performance of the AgriDrone AI system. Also included in the AgriDrone AI system interface are diagnostic ranges and classification indicators to add to the interpretability of the system.

Overall, the result from this will demonstrate that the AgriDrone AI system can accurately process unexpected input images and avoid false positive results, providing a higher level of confidence and reliability for real-world applications.

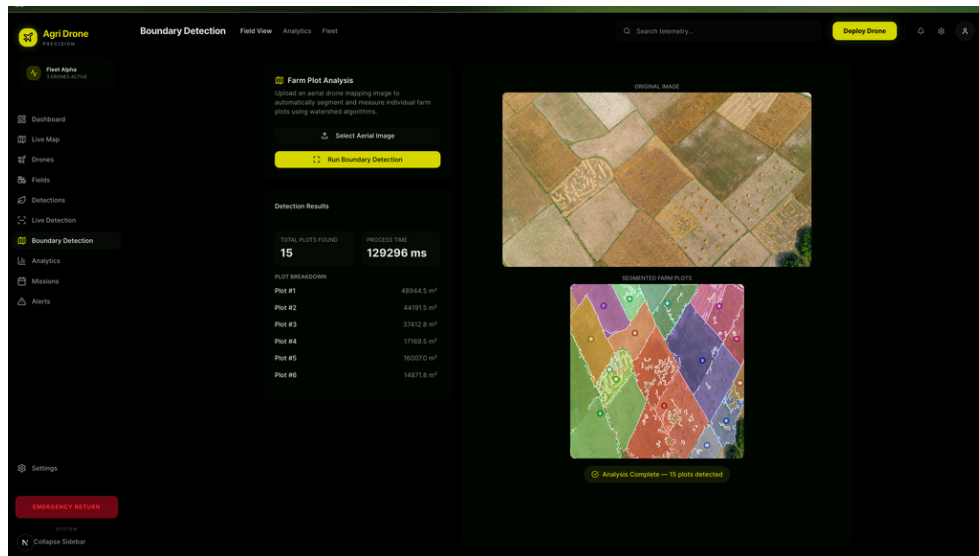


Figure 5.7: Advanced Boundary Segmentation Output Showing Complex Multi-Plot Agricultural Field

The boundary detection and segmentation results for an intricate agricultural landscape (with numerous non-regular shaped agricultural plots) are represented in Figure 5.7 (above). The top half of the figure shows the aerial photo of farmland with various texture and pattern characteristics, whilst the bottom half of the figure presents the results of segmentation.

The segmentation system has managed to separate farmland out into many separate plot categories, all having unique colors. The segmentation method has managed to accurately identify plot boundaries accurately, assessing irregular boundaries, plots with overlapping boundaries and plots with different plant density.

Metrics that were extracted from the system include: the number of plots detected, time taken for processing and the size of each plot detected.

The results displayed here were produced using a combination of edge detection and segmentation algorithms, capable of displaying field boundary definitions even under the greatest degree of complexity. The results confirm that the boundary detection module can accurately handle real-life situations when working with non-uniformly shaped fields.

The results from this figure further demonstrate that the boundary detection module is a robust and scalable solution for analysing complex farmland configurations, allowing for new understanding through spatial data.

5.11 CONCLUSION OF RESULTS

Through the experimental evaluation of the AgriDrone AI system, it has been shown that combining the use of drones as a means of collecting data with the use of AI to analyse the data collected is an appropriate method of monitoring agriculture in the twenty-first century.

The following successful objectives have been met by the system:

1. Accurate crop disease detection
2. Reliable field boundary segmentation
3. Accessible and visual representation of data

While the system has faced some minor difficulties due to environmental conditions and computing limitations, it nevertheless has tremendous potential for possible future use in real agricultural settings.

In summary; the proposed solution is a practical and scalable means of improving agricultural efficiency and will undoubtedly assist in transitioning to more precise farming techniques..

CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

AgriDrone AI provides a practical and efficient way to modernize the monitoring of agriculture through drone technology, artificial intelligence (AI), and web-based systems. This project successfully shows that combining aerial data acquisition with the latest advances in machine learning will allow us to analyze both crop health and field structure with increased accuracy and efficiency. Instead of relying solely on traditional means of manually inspecting crops, we provide a better alternative by using a drone operated manually to capture images of the crops and then applying AI based pipelines to detect disease and to segment boundaries in a field.

Using object detection to identify disease in plants and deep learning to classify diseases in plants allows us to identify potential problems with crop health before they become major problems, and therefore reduce the likelihood of crop loss and increase yield. Similarly, using an accurate field boundary detection and segmentation module gives us information regarding land distribution and area for each plot to enable more informed resource management and planning activities. By consolidating these functions into a single platform, we have significantly improved the overall usability and functionality of the system.

The system is further strengthened by its full-stack architecture, which includes a responsive frontend interface, a robust backend for processing and API management, and a structured database for data storage and retrieval. This ensures smooth communication between components and allows users to interact with the system easily while accessing meaningful insights. The modular design also makes the system scalable and adaptable for future improvements.

Although the current system operates with manual drone control and image-based analysis, it lays a strong foundation for future advancements such as autonomous drone navigation, real-time video processing, IoT-based environmental monitoring, and predictive analytics. These enhancements can further improve system capabilities and expand its applicability in precision agriculture.

In conclusion, AgriDrone AI bridges the gap between traditional farming practices and modern technological solutions by providing a data-driven approach to agricultural monitoring. It enhances efficiency, supports informed decision-making, and contributes to the development of smarter and more sustainable farming systems capable of meeting future agricultural demands.

6.2 FUTURE SCOPE

The AgriDrone AI platform provides a strong technological foundation for intelligent agricultural monitoring by integrating drone-based data collection with AI-driven analysis. Future developments will focus on enhancing automation, intelligence, and scalability while maintaining the modular architecture of the system. The following improvements can further expand its capabilities:

- **Autonomous and Semi-Autonomous Drone Operations**

The system can be upgraded to support waypoint-based navigation, automated flight paths, and obstacle avoidance. Autonomous drones can perform scheduled missions without manual control, improving efficiency and reducing human effort in large-scale monitoring.

- **Multi-Drone Coordination and Swarm Intelligence**

Future versions can include coordinated multi-drone systems that work together to cover large agricultural areas. Task allocation, collision avoidance, and synchronized data collection can significantly improve coverage speed and operational efficiency.

- **IoT Sensor Integration**

Integration with IoT devices such as soil moisture sensors, temperature sensors, humidity sensors, and nutrient detectors can provide additional environmental data. Combining this data with aerial imagery will enable more accurate and holistic analysis of crop and soil conditions.

- **AI-Based Recommendation System**

An intelligent recommendation engine can be developed to suggest appropriate pesticides, fertilizers, and irrigation strategies based on detected diseases and environmental conditions. This will help farmers take precise and timely actions.

- **Real-Time Video Processing**

Instead of relying only on static images, the system can be extended to process live video streams from the drone. This will allow continuous monitoring of crops and enable faster detection of issues in real time.

- **Predictive Analytics and Disease Forecasting**

Advanced machine learning models can be used to analyze historical data and predict future disease outbreaks or crop stress conditions. This will allow proactive decision-making and preventive measures.

- **Advanced AI Models and Ensemble Learning**

The system can incorporate multiple AI models working together (ensemble learning) to improve detection accuracy and reliability. Continuous model training using new data can further enhance performance.

- **GIS and Map-Based Integration**

Integration with Geographic Information Systems (GIS) can enable more advanced mapping, geotagging of detections, and spatial analysis of agricultural data. This will improve visualization and decision-making.

- **Mobile Application Development**

A mobile application can be developed to allow farmers to access system features on smartphones. This will improve accessibility, especially in rural areas.

- **Cloud Integration and Scalability**

Migrating the system to a cloud-based architecture will enable scalable processing, remote access, and centralized data management. It will also support real-time updates and large-scale deployment.

- **Automated Alert and Notification System**

The system can be enhanced to send real-time alerts and notifications to users when diseases or abnormalities are detected, enabling quick response and intervention.

- **Integration with Agricultural Supply Chain**

Future development can include linking the system with supply chain platforms for fertilizers, pesticides, and equipment, allowing direct action based on system recommendations.

- **Sustainability and Resource Optimization**

The system can be extended to optimize water usage, fertilizer application, and pesticide spraying, contributing to environmentally sustainable farming practices.

Overall, these future enhancements will transform AgriDrone AI into a comprehensive precision agriculture platform capable of delivering real-time monitoring, predictive insights, and intelligent decision support, ultimately improving productivity and sustainability in agriculture.

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